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Critical Review of “Timeline of Violence Since 1988”

Introduction

In this critical review, I will thoroughly assess the design quality of the "Timeline of Violence Since 1988" visualisation by Visualizing Palestine (2023) shown in Figure 1, in the context of its intended goals and underlying theoretical underpinnings. This analysis will scrutinise various aspects such as intent, influence, and impact, design and aesthetics, and usability and ethics. Each section will delve into how these elements contribute to or detract from the visualisation’s effectiveness, culminating in a nuanced understanding of its strengths and shortcomings within the broader context of data representation.

Figure 1: Timeline of Violence Since 1988

(Visualising Palestine, 2023)

Intent, Influence and Impact

Figure 1 can be understood as an explanatory visualisation, which is characterised by a deliberate stance that communicates specific patterns, trends, or discoveries that have been observed (Murray, 2015). It strives to fulfil a dual role: informing its audience of the realities of the Israeli-Palestinian conflict while simultaneously inspiring a deeper level of engagement on an emotional level—prompting more than awareness by provoking debate and reflection on the tragic scale of loss of life. Utilising data from B'selem Statistics, an award-winning Israeli organisation, the visualisation cements its credibility through stating its data source. The bar graph's description provides a narrative framework that contextualises the ongoing structural violence and its roots in what the blog terms "Israeli colonialism and apartheid" (Visualising Palestine, 2023). In doing so, the figure employs decolonial tactics as well as a feminist visualisation approach that explicitly challenges neutrality to bring to the fore aspects of the conflict that may otherwise remain unseen, particularly during the time of its creation (Amin, 2019; Visualising Palestine, 2023). This approach resonates with historical associations between gender equality and decolonisation, as outlined by McDonald (2001), where the use of force or violence, particularly in conflicts, often results in disproportionate suffering for colonised civilians, including women, children, and the elderly. The chosen theoretical approaches complement the data set and the visualisation's goals, and a bar graph is suitably selected as a medium for presenting the discrete data set effectively (Weissgerber et al., 2015), facilitating clear communication of the distinct categories and quantities and increased perceptibility (Few, 2017).

Design and Aesthetic

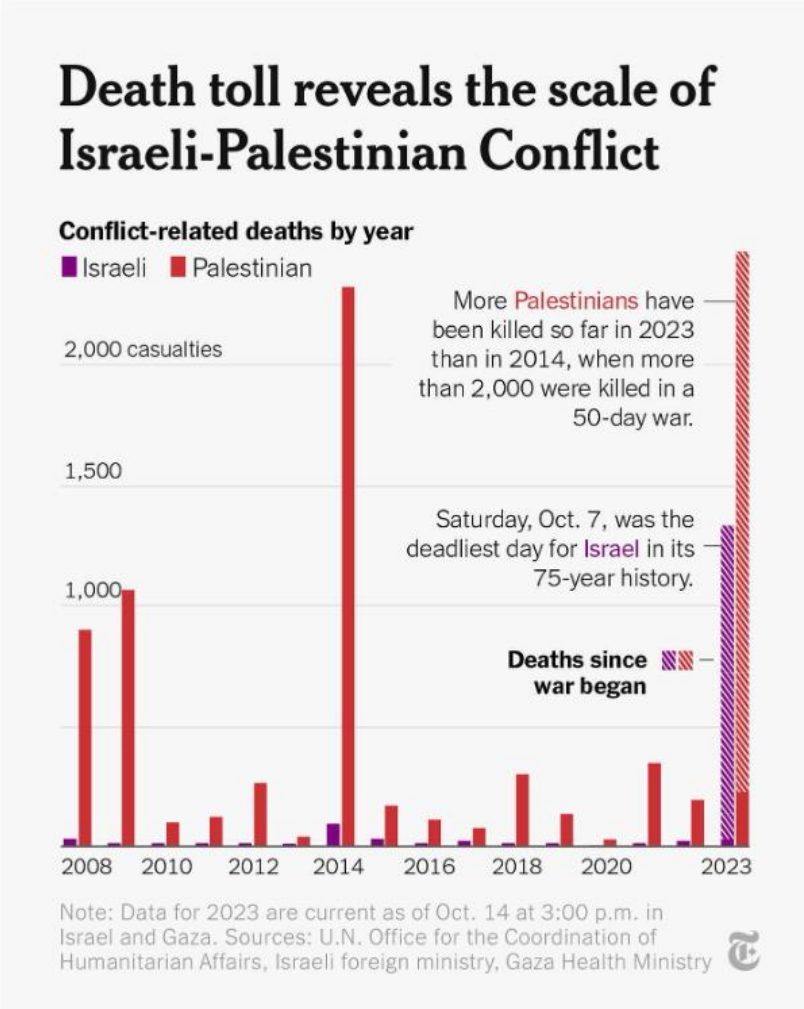
The graph's theoretical approach guides its design choices. The graph employs strategic visual highlighting- through colour, size and boldness- to guide the viewer's attention (Segel and Heer, 2010). Its use of bright red for Palestinian fatalities and black for Israeli fatalities, alongside the clear variation in bar heights, leverages pre-attentive processing to convey specific information swiftly, drawing immediate

attention to significant spikes in violence against Palestinians, like in 2014 and 2023. The graph's inclusion of large numerical labels and concise annotations provides a direct quantitative understanding of the total fatalities, effectively using negative space and typography to focus the viewer's gaze on these figures.

Usability and Ethics

However, the dual-axis approach, plotting Palestinian fatalities above the x-axis and Israeli fatalities below it, invites scrutiny. As Ware (2012) explains, feature maps in the brain are sensitive to verticality, and thus, the inverted display of Israeli fatalities could be misleading, as it may not accurately reflect the relative magnitude of the numbers presented. This is reinforced by Pandey et al. (2015), who note that directional trends—where upward movement typically signifies an increase and vice versa—can be subverted by inverting the y-axis, potentially leading viewers to draw incorrect conclusions. Furthermore, I argue that using black bars against a grey axis in the graph makes the data for some years less visible, suggesting non-existent Israeli fatalities, a misrepresentation clarified by contrasting sources such as the more neutral Figure 2 from the New York Times (which instead uses purple) and the original data source. This issue could have been rectified by annotating individual bars to show the number of fatalities each year or readjusting the scale for clarity. I suggest that it is possible and favourable to reject neutrality while simultaneously ensuring accuracy and precision by employing the suggested modifications in order to preserve the visualisation’s credibility. It is worth noting that the dual-axis approach and the use of bold contrasting colours, however, render the graph more compelling and memorable than the relatively more subdued palette utilised in Figure 2.

Figure 2: Death Toll reveals the scale of Israeli-Palestinian Conflict



(The New York Times, 2023)

I argue that the visualisation's most critical shortcoming, given its title and objective, lies in its understatement of the violence's scope by excluding non-fatal injuries. By focusing solely on fatalities, the visualisation narrows the view on violence, obscuring the wider repercussions and potentially diminishing the ongoing trauma that does not culminate in death. This exclusion may result in what Pandey et al. (2015) identify as message understatement, where the severity of the situation is downplayed through the selection of the data being visualised. Few (2017) further highlights the necessity of completeness in data visualisation, advocating that an effective depiction must present an exhaustive account by encompassing all requisite information. This encompasses not only the appropriate type and quantity of data but also the context. Absent this level of completeness, ranging from the absence of relevant data to the

inclusion of all pertinent details (through effective annotations), viewers are afforded an incomplete comprehension, potentially misconstruing the extent of the violence.

Where the visualisation excels is through its well-designed simplicity, which Few (2011) defines not merely as minimalism, but as a strategic use of design elements to effectively communicate a message without overwhelming the viewer. It avoids unnecessary 'chart junk', presenting a narrative that is clear and focused, yet it employs more than the bare minimum of ink to ensure clarity and impact. I agree with the sentiment that simplicity in design does not necessitate sparseness; it is about employing the right amount of visual cues to lead the viewer through the data meaningfully. The use of red in both figures (1 and 2) is likely deliberate, as it is often used to connotate urgency and violence, especially in contexts of war (Yonder, 2022). Unlike Figure 2, the utilisation of a dual-axis in this visualisation allows viewers to easily discern which bars represent which data set, effectively mitigating any potential confusion arising from the inaccessible colour palette, despite the conventional guidance to avoid red for better accessibility (Katnelson, 2021).

Conclusion

Figure 1 offers a compelling narrative through its strategic design choices and theoretical approaches, despite its shortcomings. Its emphasis on fatalities in the Israeli-Palestinian conflict effectively communicates the scale of loss while its dual-axis design and colour scheme draw attention to critical disparities. However, the visualisation's limited scope, potentially misleading graphical elements, and understatement of non-fatal violence highlight the challenges in balancing impact with accuracy and completeness. This review underscores the importance of a holistic approach in data visualisation, where design, ethics, and context converge to present a narrative that is both informative and memorable.

References:

Amin, B. (2019) Data Visualization: Building A New Narrative of Palestine for the Digital Age, *SceneArabia*. Available at: http://www.SceneArabia.com/Life/visualizing-palestine-israel-data-visualisation-war-on-narrative (Accessed: 10 November 2023).

B'Tselem. (2023) Statistics. Available at: https://www.btselem.org/statistics (Accessed: 14 November 2023).

Few, S. and Edge, P., 2011. The chartjunk debate. *Visual Business Intelligence Newsletter*, no. June, pp.1-11.

Few, S. (2017). Data Visualization Effectiveness Profile. Available at: https://www.perceptualedge.com/articles/visual_business_intelligence/data_visualization_effectiveness_profile.pdf (Accessed: 9 November 2023).

Katnelson, A. (2021) Colour me better: fixing figures for colour blindness, *nature*. Available at: https://www.nature.com/articles/d41586-021-02696-z (Accessed: 9 November 2023).

McDonald, S., 2001. Kurdish women and self-determination: A feminist approach to international law. In: *Women of a non-state nation: The Kurds*, 7, pp.135-157.

Murray, S.A. (2015). Mapping the Spectrum of Intent. In: Bihanic, D. (ed.), *Data Visualization and Data Arts. In New Challenges for Data Design*. Springer.

Pandey, A.V., Rall, K., Satterthwaite, M.L., Nov, O., Bertini, E. (2015). How Deceptive are Deceptive Visualizations? An Empirical Analysis of Common Distortion Techniques. *Proceedings CHI '15 Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, pp. 1469-1478. Available at: https://dl.acm.org/doi/10.1145/2702123.2702608 (Accessed: 9 November 2023).

Segel, E. and Heer, J. (2010) 'Narrative Visualization: Telling Stories with Data', *IEEE Transactions on Visualization and Computer Graphics*, 16(6), pp. 1139–1148. Available at: https://doi.org/10.1109/TVCG.2010.179.

The New York Times (2023) Death Toll reveals the scale of Israeli-Palestinian Conflict. *Instagram*. Available at: https://www.instagram.com/p/CyZOFjvOmmk/ (Accessed: 8 November 2023).

Visualising Palestine (2023) Timeline of Violence-October 2023. Available at: https://www.visualizingpalestine.org/visuals/timeline-violence-2023#&gid=1&pid=1 (Accessed: 14 November 2023).

Ware, C. (2012). Chapter 5 – Visual Salience and Finding Information. In: Ware, C. (ed.), *Information Visualisation: Perception for Design*. Morgan Kauffman.

Weissgerber, T.L., Milic, N.M., Winham, S.J. and Garovic, V.D., 2015. Beyond bar and line graphs: time for a new data presentation paradigm. *PLoS biology*, 13(4), p.e1002128.

Yoder, D., 2022. Data: What's the Point? How Curiosity Can Help Us Put Information in Context. *Society and Culture*.